

Appendix: Project Deal

April 24, 2026

This appendix documents the data, estimation approach, and full regression results behind the statistical claims in the main text. Section A summarizes the experimental design. Section B presents summary statistics. Sections C–H present full specifications and regression tables for each outcome.

A Experimental Design

A.1 Run structure

The experiment comprised four independent, simultaneous marketplace runs. Each run involved the same 69 participants listing the same items. The runs differed in two dimensions: agent-model assignment and visibility on company Slack.

Table 1: Factorial design of marketplace runs.

	All Opus 4.5	50% Opus 4.5 / 50% Haiku 4.5 (randomized)
Public	A	B
Private	D	C

Runs A and D are *symmetric* markets (all agents are Opus). Runs B and C are *asymmetric* markets (mixed agent quality). Agent assignment in B and C was randomized independently across the two runs; a given participant could receive Opus in B and Haiku in C or vice versa.

All of our Opus-vs-Haiku estimates come from variation in Runs B and C alone as Runs A and D provide no within-run variation (every participant had Opus). In our main specifications, we only rely on B and C runs to estimate the model. The exception is the joint within-item price model (§E.2), where runs A and D are used to anchor item fixed effects but do not add treatment variation after run fixed effects are partialled out.

A.2 Timeline

- **Onboarding interviews.** Claude interviewed each participant, eliciting items for sale, purchasing preferences, price expectations, and negotiating-style instructions. Each interview produced a custom system prompt for that participant’s agent.
- **Market runs.** Four simultaneous runs operated for approximately one week.
- **Survey.** Exit survey administered after market close. Participants did not know which run was “real” until after they completed the survey.
- **Exchange event.** Participants exchanged the physical goods agreed to in run A.

A.3 Participants and incentives

Sixty-nine Anthropic employees in the San Francisco office volunteered. Each received a \$100 notional budget; final payout equaled \$100 plus the net value of completed transactions (items sold minus items bought at agreed prices), delivered as a gift card. The payout mechanism was used largely to incentivize participation.

B Summary Statistics

B.1 Participation

Table 2: Participation counts.

	N
Total participants	69
Participants who listed items	69
Exit surveys started	61
Surveys reaching the run-ranking section	61
Surveys reaching the willingness-to-pay section	52

B.2 Market activity by run

Each run has the same 575 listed items across the same 69 sellers. Sale prices are conditional on sale.

Table 3: Market activity by run.

Run	Model assignment	Items listed	Items sold	Sale rate	Mean price	Total value
A	All Opus	575	206	35.8%	\$19.46	\$4,010
B	Mixed (Opus/Haiku)	575	167	29.0%	\$20.38	\$3,404
C	Mixed (Opus/Haiku)	575	197	34.3%	\$20.15	\$3,970
D	All Opus	575	212	36.9%	\$20.25	\$4,292
Total		2,300	782	34.0%	\$20.05	\$15,676

B.3 Market activity by seller agent type (Runs B and C only)

Table 4: Market activity by seller agent type, Runs B and C.

Seller agent	Items listed	Items sold	Sale rate	Mean price (if sold)
Opus	619	208	33.6%	\$21.11
Haiku	531	156	29.4%	\$19.12

B.4 Price distribution (all sold items, all runs)

Table 5: Price distribution across all completed transactions.

Statistic	Value
N (completed transactions)	782
Mean	\$20.05
Median	\$12.00
Standard deviation	\$20.86
Minimum	\$0.00
Maximum	\$175.00

The large mean–median gap reflects a right-skewed distribution—most items are inexpensive everyday goods, with a long tail of higher-value items.

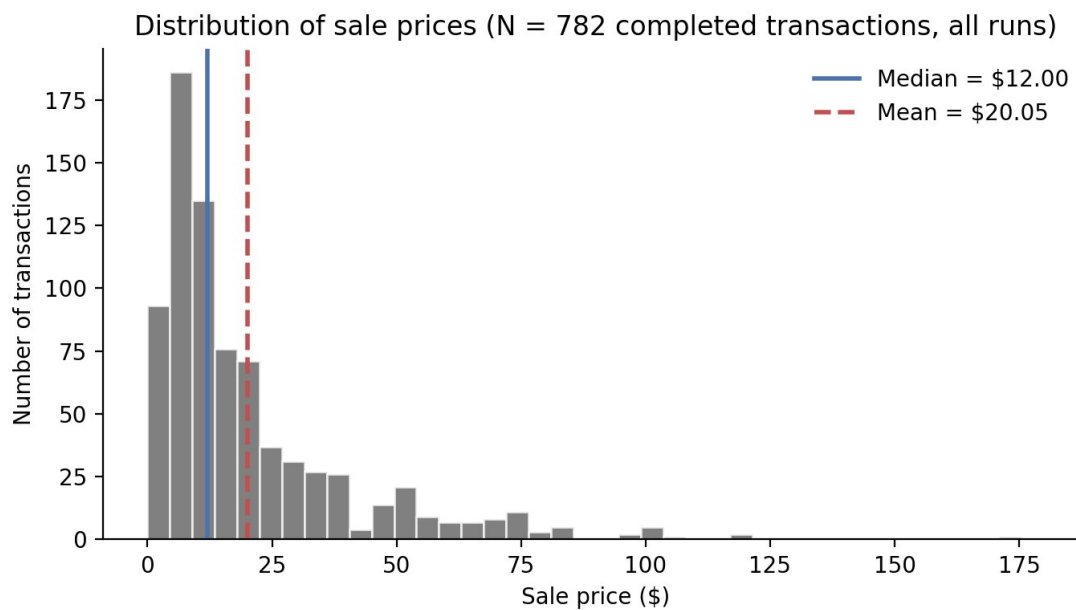


Figure 1: Histogram of sale prices (all 782 transactions), with median and mean annotated.

B.5 Transaction prices by buyer–seller matchup (Runs B and C)

Each cell shows the mean sale price and number of transactions for each combination of seller model and buyer model.

Table 6: Mean transaction prices by buyer–seller matchup, Runs B and C.

	Opus buyer	Haiku buyer
Opus seller	\$18.63 (n=115)	\$24.18 (n=93)
Haiku seller	\$18.84 (n=104)	\$19.69 (n=52)

B.6 Prompting-style classification

Claude read each participant’s onboarding interview transcript and coded it, blind to experimental outcomes, to determine if the agent was prompted to be *aggressive* (the participant instructed their agent to negotiate aggressively or hard-bargain).

Table 7: Prompting-style classification of participants.

Prompt style	N participants	Share
Aggressive	28	40.6%
Non-aggressive	41	59.4%

C Deal Volume (Person–Run)

C.1 Sample

Person–run panel from Runs B and C only. One observation per person per run, 61 persons $\times$ 2 runs = 122 observations. Of the 61 persons, 28 had different agents across the two runs (one Opus, one Haiku) and 33 had the same agent in both.

C.2 Specification

$$\text{DealCount}_{ir} = \alpha + \beta \cdot \text{Opus}_{ir} + \gamma_i + \delta_r + \varepsilon_{ir} \quad (1)$$

where i indexes persons, $r \in \{B, C\}$ indexes runs, $\text{Opus}_{ir} \in \{0, 1\}$ indicates Opus assignment, γ_i is a person fixed effect, and δ_r is a run fixed effect (included in (D) as a robustness check). Standard errors clustered by person.

γ_i absorbs everything constant about a participant across the two runs such as number and desirability of listed items and purchasing preferences. β is therefore identified from the 28 participants with differing agent assignments; the 33 “non-switchers” are absorbed by their person fixed effect.

C.3 Results

Table 8: Deal-volume regressions. Dependent variable: completed deals in the run.

	(A) OLS, clustered SE	(B) Person FE	(C) Person FE + Run FE
Opus	1.121** (0.542)	2.071*** (0.596)	2.110*** (0.568)
Person FE	No	Yes	Yes
Run FE	No	No	Yes
Clustered SE (person)	Yes	Yes	Yes
N (observations)	122	122	122
N (clusters)	61	61	61

Raw means: Opus 5.33 deals per person-run (n=62); Haiku 4.21 (n=60).

D Item Sale Probability (Item–Run)

D.1 Sample

Item–run observations from runs B and C only: 1,150 observations ($575 \text{ items} \times 2 \text{ runs}$) (see §A.1).

D.2 Specification

$$\text{Sold}_{isr} = \alpha + \beta \cdot \text{OpusSeller}_{sr} + \gamma_s + \delta_r + \varepsilon_{isr} \quad (2)$$

where i indexes items, s indexes sellers (fixed for each item), $r \in \{B, C\}$ indexes runs, $\text{OpusSeller}_{sr} \in \{0, 1\}$ indicates that seller s had Opus in run r , γ_s is a seller fixed effect, and δ_r is a run fixed effect. Standard errors clustered by seller ($G = 69$).

γ_s absorbs everything constant about a seller across the two runs (items offered, minimum acceptable prices, general desirability as a counterparty). Identification comes entirely from the random agent swap between B and C.

D.3 Results

Table 9: Item sale probability regressions. Linear probability model; coefficients in percentage points.

	(1) Seller FE	(2) Seller FE + Run FE
Opus seller	+5.58 pp (3.51 pp)	+6.63 pp* (3.43 pp)
Seller FE	Yes	Yes
Run FE	No	Yes
Clustered SE (seller)	Yes	Yes
N (observations)	1,150	1,150
N (sellers / clusters)	69	69

Raw sale rates in Runs B and C: Opus sellers 33.6%, Haiku sellers 29.4%.

E Within-Item Price Effects

E.1 Paired seller-price comparison

Sample. The 44 items that sold in both Run B and Run C *and* whose seller was assigned different models in the two runs.

Test. For each qualifying item j , compute

$$\Delta_j = p_j(\text{Opus seller}) - p_j(\text{Haiku seller}) \quad (3)$$

and conduct a one-sample t -test of $H_0 : \mathbb{E}[\Delta_j] = 0$.

Result. $\bar{\Delta} = +\$3.64$, $t(43) = 2.644$, $p = 0.011$. Opus mean \$25.11; Haiku mean \$21.47.

E.2 Joint item-fixed-effects price model

Sample. All 782 sold item–run observations across all four runs; 575 unique items; 161 items with within-item variation in buyer or seller model.

Specification.

$$p_{ijr} = \alpha + \beta_1 \cdot \text{OpusBuyer}_{jr} + \beta_2 \cdot \text{OpusSeller}_{ir} + \gamma_j + \delta_r + \varepsilon_{ijr} \quad (4)$$

where p_{ijr} is the sale price of item j sold by seller i in run r , γ_j is an item fixed effect, and δ_r is a run fixed effect. Standard errors clustered by seller ($G = 69$).

Results.

Table 10: Joint within-item price model. Dependent variable: sale price (USD).

(1) All four runs	
Opus buyer	−\$2.45** (\$0.98)
Opus seller	+\$2.68** (\$1.21)
Item FE	Yes
Run FE	Yes
Clustered SE (seller)	Yes
N (transactions)	782
N (items with variation)	161
N (seller clusters)	69

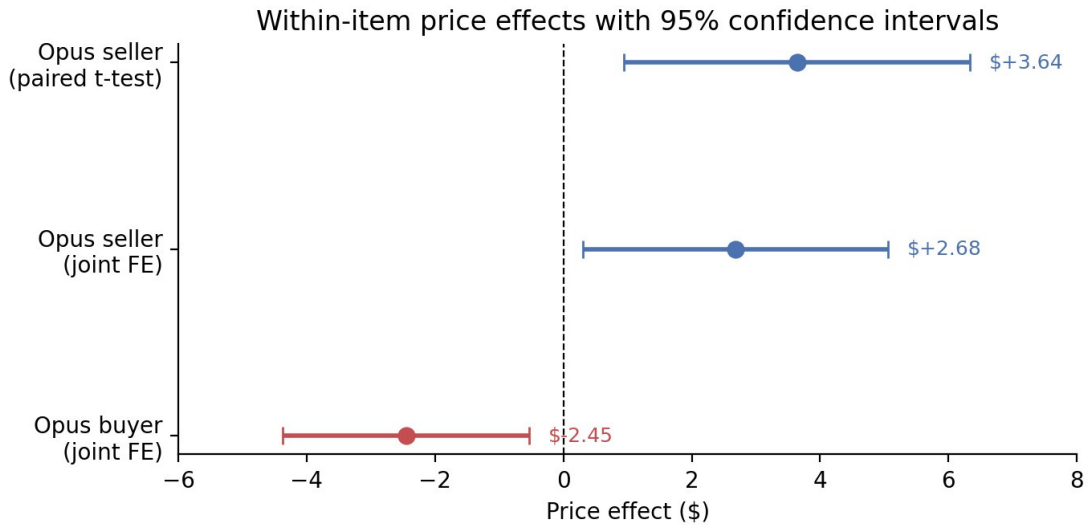


Figure 2: Forest plot of the three price effects—paired Opus-vs-Haiku seller effect, joint Opus-buyer coefficient, joint seller-Opus coefficient—with 95% confidence intervals.

F Deal Satisfaction and Fairness (Deal Level)

F.1 Sample

Deal-level observations from Runs B and C: 585 deals, 61 unique persons (341 deals by Opus-assigned users, 244 by Haiku-assigned users). Twenty-six persons had deals under both agent types and contribute within-person variation.

F.2 Specification

$$\text{Satisfaction}_{idr} = \alpha + \beta \cdot \text{Opus}_{ir} + \gamma_i + \delta_r + \varepsilon_{idr} \quad (5)$$

where i indexes persons, d indexes individual deals, $r \in \{B, C\}$, γ_i is a person fixed effect. Standard errors clustered by person.

F.3 Satisfaction results

Table 11: Deal satisfaction regressions. Dependent variable: satisfaction on a 1–7 scale.

	(A) Clustered	(B) Person FE	(C) +Run FE	(D) +Run FE +Order
Opus	+0.311 (0.214)	+0.217 (0.244)	+0.200 (0.246)	+0.136 (0.248)
Person FE	No	Yes	Yes	Yes
Run FE	No	No	Yes	Yes
Survey order	No	No	No	Yes
Clustered SE	Yes	Yes	Yes	Yes
N (deals)	585	585	585	585
N (clusters)	61	61	61	61

Raw means: Opus 5.14; Haiku 4.82.

F.4 Fairness results

Fairness was rated on the same 1–7 scale (1 = unfair to buyer, 4 = fair to both, 7 = unfair to seller).

Table 12: Deal fairness ratings by agent type.

	Opus	Haiku	Difference	p (naive)	p (clustered)	p (person FE)
Mean fairness	4.053	4.045	+0.008	0.921	0.943	0.339

Fairness is indistinguishable across agent types in every specification. Not only do participants not perceive a satisfaction gap, they also do not perceive Haiku deals as systematically *unfair* to them, even though Haiku buyers in fact paid measurably more and Haiku sellers in fact received less.

G Run Rankings

G.1 Design

At the end of the survey, each participant ranked the four runs’ bundles of outcomes from best (1) to worst (4).

G.2 Mean ranks by run

Lower rank = preferred.

Table 13: Mean rank by run (lower is preferred).

	All Opus	Mixed
Public	2.44 (Run A)	2.46 (Run B)
Private	2.54 (Run D)	2.56 (Run C)

A Friedman test on within-person rankings across the four runs does not reject the null of identical distributions.

G.3 Within-person sign test (mixed-agent participants)

Of the 61 respondents, 28 had different agents in Runs B and C (one Opus run, one Haiku run).

Table 14: Within-person sign test, mixed-agent participants.

	N	Share
Ranked Haiku run worse (higher number)	17	60.7%
Ranked Haiku run better (lower number)	11	39.3%
Tie	0	—

Two-sided binomial sign test of $H_0 : \Pr(\text{Haiku ranked worse}) = 0.5$:

$$p = 2 \sum_{k=17}^{28} \binom{28}{k} (0.5)^{28} = 0.345 \quad (6)$$

A paired t -test on the within-person rank differences gives mean +0.50 rank positions, $t(27) = 1.52$, $p = 0.143$ —directionally consistent with a Haiku disadvantage but not significant.

H Prompting Style (Aggression)

H.1 Coding

As noted above, we used Claude to read participants’ onboarding interview transcripts and classify them as *aggressive* (the participant instructed their agent to negotiate assertively or hard-bargain) or not.

Classification: 28 of 69 (40.6%) aggressive; 41 of 69 (59.4%) non-aggressive.

H.2 Specifications

Model A — Sale rate (item–run level):

$$\text{Sold}_{isr} = \alpha + \beta \cdot \text{AggSeller}_s + \delta_r + \varepsilon_{isr} \quad (7)$$

Seller aggression is invariant, so seller fixed effects cannot be included. Standard errors clustered by seller.

Model B — Price, buyer aggression (transaction level):

$$p_{ijr} = \alpha + \beta \cdot \text{AggBuyer}_j + \gamma_s + \delta_r + \varepsilon_{ijr} \quad (8)$$

Seller fixed effects γ_s absorb all item-composition variation across sellers. β is identified from within-seller variation in buyer aggressiveness across transactions. Standard errors clustered by seller.

Model C — Price, seller aggression (transaction level):

$$p_{ijr} = \alpha + \beta \cdot \text{AggSeller}_s + \gamma_j + \delta_r + \varepsilon_{ijr} \quad (9)$$

Buyer fixed effects γ_j absorb fixed traits of the buyer. β is identified from within-buyer variation in seller aggressiveness across purchases (57 of 69 buyers had variation). Standard errors clustered by seller.

H.3 Results

Table 15: Prompting-style (aggression) regressions.

	Model A (sale rate)	Model B (price, buyer agg)	Model C (price, seller agg)
<i>All four runs</i>			
Aggressive	+5.2 pp (6.6 pp)	+\$0.56 (\$1.99)	+\$6.04* (\$3.05)
<i>p</i>	0.428	0.778	0.052
<i>Runs B and C only</i>			
Aggressive	+8.7 pp (6.6 pp)	+\$0.15 (\$2.74)	+\$3.84 (\$3.40)
<i>p</i>	0.192	0.957	0.263
Seller FE	No	Yes	No
Buyer FE	No	No	Yes
Run FE	Yes	Yes	Yes
Clustered SE (seller)	Yes	Yes	Yes
N (all four runs)	2,300	782	782
N (B and C only)	1,150	364	364

Model A uses all item–run observations (listed items, sold or not). Models B and C are estimated on sold items only.

Interpretation of Model B. Aggressively-prompted buyers did not pay less. If anything the point estimate is (non-significantly) positive.

Interpretation of Model C. The raw Model C estimate of +\$6.04 ($p = 0.052$) primarily reflects the fact that aggressive-instruction sellers stated higher asking prices in their onboarding interviews (about \$26 more per item on average)—not that their agents negotiated harder once bargaining began. Replacing

the dependent variable with the *fraction of the asking-to-minimum-price spread* captured by the seller, $(p - p_{\min}) / (p_{\text{ask}} - p_{\min})$, nets out the asking-price channel and asks whether aggressive prompting produced better negotiated outcomes conditional on the stated spread. See §H.4.

H.4 Model D — Spread share captured by the seller

Model C measures the effect of aggressive prompting on the raw sale price, which confounds two channels: (1) aggressive sellers named higher asking prices up front, and (2) aggressive sellers’ agents may have negotiated harder once bargaining began. To isolate channel (2), we replace the dependent variable with the share of the asking-to-minimum-price bargaining spread that the seller captured. Each seller stated both an asking price p_i^{ask} and a minimum acceptable price $p_i^{\min}$ during their onboarding interview. The share captured is

$$s_{ijr} = \frac{p_{ijr} - p_i^{\min}}{p_i^{\text{ask}} - p_i^{\min}} \quad (10)$$

where $s = 1$ means the seller got their full asking price and $s = 0$ means they sold at their minimum. We clip s at 1.5 to handle the small number of cases where the agent sold above the human-stated ask without letting those cases dominate the estimate.

Specification.

$$s_{ijr} = \alpha + \beta \cdot \text{AggSeller}_s + \gamma_j + \delta_r + \varepsilon_{ijr} \quad (11)$$

Buyer fixed effects γ_j absorb all buyer-level differences in willingness to concede; the run fixed effect absorbs any run-wide shift in how much of the spread sellers typically captured. Standard errors clustered by seller. Estimated on sold items with a strictly positive ask–min spread.

The coefficient β measures the fraction of potential seller surplus that an aggressive seller ultimately captures. For comparability with Model C we also report a dollar-equivalent effect computed as $\hat{\beta} \times \bar{S}_{\text{non-agg}}$, where $\bar{S}_{\text{non-agg}} = \8.35 is the mean ask–min spread across items offered by non-aggressive-instruction sellers. This is the counterfactual dollar gain if non-aggressive sellers retained their stated spreads but their agents captured shares at the aggressive rate; alternative reference spreads are reported below as a sensitivity check.

Results.

Table 16: Model D: spread share captured by the seller.

	(1) Share (native units)	(2) Dollar equivalent at $\bar{S}_{\text{non-agg}} = \8.35
Aggressive seller	+0.118 (0.107)	+\$0.95 (\$0.89)
Buyer FE	Yes	Yes
Run FE	Yes	Yes
Clustered SE (seller)	Yes	Yes
N (transactions)	453	453
N (seller clusters)	53	53
p	0.275	0.275

Sensitivity of the dollar translation to the reference spread.

Table 17: Sensitivity of the dollar-equivalent effect to choice of reference spread.

Reference spread	Value	Dollar-equivalent effect
Non-aggressive-seller median	\$5.00	+\$0.57
Non-aggressive-seller mean (preferred)	\$8.35	+\$0.95
Overall median	\$5.00	+\$0.57
Overall mean	\$11.66	+\$1.33

Interpretation. Once the asking-price channel is netted out, the residual negotiation-performance effect of aggressive prompting is roughly a dollar per item and not statistically distinguishable from zero. Almost the entire raw dollar gap in Model C traces to aggressive-instruction sellers having asked for more at the outset, not to their agents extracting more during bargaining. We do not interpret a causal effect of aggressive prompting on prices.